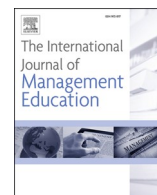




Contents lists available at ScienceDirect

The International Journal of Management Education

journal homepage: www.elsevier.com/locate/ijme

Transformative learning in business economics education: The development of a measurement instrument to detect student perspectives

Lise Janssens^{a,f,*}, Tom Kuppens^{a,b,c,f}, Pernilla Andersson^{d,e}, Katrien Struyven^b^a UHasselt, Centre for Environmental Sciences, Research Group Environmental Economics, Agoralaan, 3590, Diepenbeek, Belgium^b UHasselt, School of Educational Studies – URISSE UHasselt Research on Innovative and Society-engaged Education, Wetenschapspark 42, 3590, Diepenbeek, Belgium^c Vrije Universiteit Brussel, Multidisciplinary Institute Teacher Education (MILO), Pleinlaan 9, 1050, Elsene, Belgium^d Stockholm University, Department of Teaching and Learning, 106 91, Stockholm, Sweden^e Uppsala University, Department of Education, Box 2136, SE-750 02, Uppsala, Sweden^f COPERNICUS Alliance - European Network on Higher Education for Sustainable Development, Sweden

ARTICLE INFO

Keywords:

Sustainability education
Business economics education
Transformative learning
Scale development
Responsibility in business
Student perceptions

ABSTRACT

Previous studies of business economics education have highlighted differences regarding the role of businesses towards sustainability, as articulated by teachers, students and in textbooks. The roles differ from externally motivated (following law and customers) to internally driven (creating new routes) to address sustainability in the business context. Changes by students in their perspectives on the role of a business in relation to sustainability can be seen as examples of a transformative learning experience in the context of business economics education. This paper presents a measurement instrument of 12 items on a five-point Likert scale to measure the views of students regarding the role of a responsible businessperson. A mixed-methods approach was used to develop, test and validate the instrument. Data were gathered by business economics students at Hasselt University. The added value of the instrument is that it can be used in educational practice and in further research to indicate whether transformative learning is happening in the context of business education and to gain insight into students' views on the role of a responsible businessperson. The instrument goes beyond existing attitude, knowledge and competence scales and contributes to assessing transformative learning for sustainable development.

1. Introduction

Sustainability education has been described as a key element in transforming to a sustainable society (Howlett et al., 2016; Gómez et al., 2015; Wiek et al., 2013), both in higher education institutions (HEIs) (Howlett et al., 2016) and secondary schools (Kane, 2011). To overcome the challenges related to sustainable development, the European Commission proposed, with GreenComp, four competence domains, including 'embodying sustainability values' to support sustainable development (Bianchi et al., 2022).

* Corresponding author. UHasselt, Centre for Environmental Sciences, Research Group Environmental Economics, Agoralaan, 3590, Diepenbeek, Belgium.

E-mail address: lise.janssens@uhasselt.be (L. Janssens).

<https://doi.org/10.1016/j.ijme.2024.100988>

Received 19 October 2023; Received in revised form 13 March 2024; Accepted 18 April 2024

Available online 2 May 2024

1472-8117/© 2024 Elsevier Ltd. All rights reserved.

Moreover, to encourage more sustainable behavior, Sidiropoulos (2014) proposed including values as an important learning outcome when implementing Education for Sustainability (EfS) in tertiary business education programmes. The approach of transformative learning for sustainability education has been encouraged by UNESCO as a way of transitioning from unsustainability to more sustainability (UNESCO, 2021). Sustainability is a process and no predefined solutions exist. Therefore, instrumental and linear methods in sustainability education are problematic because they might imply the existence of a single solution for sustainability (Lambrechts, Van Liedekerke, and Van Petegem 2018). Using a transformative learning approach overcomes the paradox of using pedagogies focused on knowledge transfer to teach a curriculum that has the potential to transform the thinking and actions of students (Seatter & Kim, 2017). Transformative learning “(...) involves a deep structural shift in the basic premises of thought, feelings and actions. It is a shift of consciousness that dramatically and permanently alters our way of being in the world” (Morrell & O’Connor, 2002, p. 17). Some instruments to assess transformative learning process exist, such as ‘Kember’s Critical Reflection Questionnaire’, the ‘Learning Activity Survey’, the ‘Transformative Learning Survey’ and the ‘VALUE rubric’. Mostly, these instruments combine qualitative and quantitative instruments to measure parts of the transformative learning process, such as critical thinking (Romano, 2018). However, the approach of transformative learning is challenging to assess because the degree of learning depends on the quality of the learning environment and the openness of the learner and is highly personal (Sterling, 2011). Therefore, assessing transformative learning in practice is difficult (Morrell and O’Connor (2002). The existing measurement tools are all very general, not context-specific and do not fit to the context of business economics education. With this background, the paper presents a self-report questionnaire to identify students’ different perspectives on the role of a business in relation to sustainability. Currently, no instruments exist to detect the different roles of a responsible businessperson internalised by students and teachers. Based on textbook analyses, classroom observations and interviews, Andersson (2016) argued that this moral driver to act for sustainability could be internal or external. Three different roles of a responsible businessperson were described based on different moral drivers to act for sustainability. Also, Raworth (2017) described different approaches to addressing sustainability in business. A distinction has been made among (1) doing nothing except what is required by law, (2) adopting eco-efficient practices to reduce costs or to create a good reputation, and (3) not causing any harm, or even having a positive impact on the environment. Changes in the roles of a responsible businessperson or approaches about why to integrate sustainability in business – for example, during a course – can be seen as an example of a transformative learning experience by the student in the business economics class. In this way, the instrument contributes to assessing transformative learning in the business economics education context.

Creating leaders for the 21st century is one of the major responsibilities of business schools and colleges (McFarlane & Ogazon, 2011). Business economics students need to be able to address the challenges of the 2050s. However, the handbooks used in their education are from the 1950s representing theories developed in the 1850s (Raworth, 2017). Future (business) leaders have to be able to employ sustainability concepts, implement corporate social responsibility in the organisation, and be able to work with stakeholders with different interests (Annan-Diab and Molinari 2017). Although these have been discussed in the past (see, for example, Horton, 1972), there is an increased focus on integrating ethics, sustainability and responsibility in the curriculum of business economics students (Colander, 2017; Stough, Ceulemans, and Cappuyns 2021). Integrating sustainability and responsibilities in business education is necessary in order to face social, economic and environmental challenges such as poverty and resource scarcity. In addition, businesses are increasingly interested in sustainability competences by their future employees because they want to transform their business into more sustainable alternatives (Baumgartner, 2014). Various reasons cause this interest, such as intrinsic motivation, customer demands (Harris, 2007), possible financial benefits, overcoming problems of resource scarcity or legal concerns (Saha, 2014). Initiatives taken by the United Nations, such as The Principles for Responsible Management Education (PRME), aim to transform management education, research and leadership and to promote the adoption of the sustainable development goals (SDGs) (Storey et al., 2017). The PRME provide a vision for how management education should be and introduce a way of thinking about education in business schools (Millar & Price, 2018). Research studies focusing on the integration of sustainability in business education have reflected the increased attention given to the topic (Beddewela et al., 2017; Greenberg et al., 2017; Kolb et al., 2017). Also, students’ interest in this topic is growing (Maxfield, 2011). Integrating sustainability in business economics education is important because economists often play a central role in government organisations and companies. Business economics students are trained to make choices due to scarcity (Backhouse & Medema, 2009). In their future jobs they often need to make decisions that influence the policy of their organisation, either directly or indirectly. In this way, by making decisions they can influence a transition to sustainable development (Ceulemans et al., 2011). As economic decisions can have profound environmental and social impacts, it is important to implement ‘sustainability’ in the decision-making process (Green, 2015). Unfortunately, in the training of business economics students, the principle of the *homo economicus* is still central (Raworth, 2017), self-interest is encouraged (Wang et al., 2011), and simplified and unrealistic representations of the world are used (Birks, 2012). The principle of scarcity is dominant in business economics education, even though it is not always valid or desirable (Darity, 2022). Business economics education has also failed to anticipate crises such as the 2007–2008 financial crisis (Colander et al., 2009). Movements like rethinking economics have called for changes to economics education and integrate different perspectives in (business) economics education (de Muijnck & Tieleman, 2022). So, although the need to integrate sustainability into (business) economic courses is no longer questioned, its uptake in recent years has fallen behind expectations (García-Feijoo et al., 2020). Initiatives such as the PRME have created awareness about the necessity of integrating responsibilities in business but little space is left for critical reflection to question the fundamental understandings and practices in the field (Millar & Price, 2018). It is important to integrate sustainability in business economics education to stimulate future economists to take sustainability into account in their professional work without defining what they should do and how they should react.

Integrating sustainability in the field of business economics education is challenging because it involves integrating continuously changing individual and social values (Sidiropoulos, 2014). Different perspectives on the role of a business in relation to sustainability

exist, as illustrated in the research of Andersson (2018) and in the book *Doughnut Economics* by Raworth (2017). The different perspectives reflect different *motivations* behind their actions although the actions can be the same. In her work, Andersson (2018) explored how different discourses – that is, the assumptions and contentions with which people look at and talk about the world, regarding the ethical responsibilities of businesses – influence the way sustainable development is understood by teachers. Also, García-Rosell (2019) argued that the discourse of how the responsibilities of a business are seen in the wider economic and social system influences the framing of sustainability and sustainability education. Andersson (2016) identified the roles of a responsible business as articulated in (business) economics textbooks and by teachers as adapting to, adding or creating ethical values. A businessperson was defined as anyone making a business decision, such as an owner or a manager. The adapting role implies that the businessperson is expected to maximise profit while following the law. Teachers and textbooks adopting the adapting role have stated that personal convictions should not be considered and that business decisions are considered responsible as long they do not infringe on rules and legislation. The adding role implies that a businessperson is expected to do more than what is required by law and regulations, fulfilling the needs of people external to the business. ‘Responsible’ in this discourse is seen as fulfilling customers’ demands. According to the creating role, the businessperson is guided by an internal desire for the business to make change, to be responsible, and to be sensitive to the needs of all when making a business decision. Teachers and textbooks adopting the creating role state that the values and feelings of the businessperson play a key role when making a business decision. In her book, Raworth (2017) presented a visual framework, shaped as a doughnut, for sustainable development. The area within the doughnut is a safe space for humanity where environmental and social boundaries are considered. Based on Raworth’s framework, doughnut economics is a theory that proposes change in the current economic model to overcome the global challenges. In her book, Raworth described different approaches to doing business sustainably. The different approaches can be summarised as (1) doing nothing except what is requested by law and legislation, (2) taking eco-efficient measures to lower costs or to create a good reputation, or (3) delivering a fair share to sustainability, not causing any harm or even aiming to have a positive impact on the environment.

The different perspectives on how to do sustainability within business can be compared to more general literature about responsibilities. Pellizzoni (2004) developed a typology of responsibilities to illuminate strengths and weaknesses in relation to complex sustainability issues; namely, (1) liability regime, (2) accountability regime, (3) responsiveness regime and (4) care regime. In a liability regime, good governance, rules, laws and written agreements are central. Responsibilities are defined by what is required by rules and legislation. ‘Good’ behavior is created by legal sanctioning (see also Faure, 2009, p. 247). The liability regime can be linked to the adapting role of Andersson (2016) or doing nothing except what is required by law, as described by Raworth (2017) specifically in the context of sustainability in business. In an accountability regime, governance is central; however, in contrast with the liability regime, this is based on voluntary actions. In that sense, responsibilities for sustainability in the business context are dependent on what the customers ask for; for instance, through buying eco-labelled products, as illustrated in the adding role of Andersson (2016), or taking eco-efficient measures as described by Raworth (2017). In a responsiveness regime, the moral impetus to act comes from the inside and is not driven by an external request (Pellizzoni, 2004). Related to a business context, this means a business is sensitive to the needs of stakeholders and socio-ecological systems. The guiding principle for making a sustainable business decision is internal, as defined in the creating role of Andersson (2016) and described by Raworth (2017) as delivering a fair share to sustainability causing no harm. Although not found in the work of Andersson (2016) and Raworth (2017), Pellizzoni (2004) described a fourth responsibility regime: care. In a care regime, taking care of others is central. A possible reason for the absence of a link to the care regime in the studies of Andersson (2016) and Raworth (2017) can stem from the normative nature of the care regime, suggesting that it is known what is best. Andersson (2016), Raworth (2017) and Pellizzoni (2004) agreed that creating new paths to sustainability and act responsibly are important and preferable in order to address complex sustainability issues. Different reasons can be enumerated for this argument. The most important one is that following authority, following law or following customers’ demand will only result in sustainable actions if ‘the law is good enough’ and customers ‘ask for decent changes in regard to sustainability.’

A lot of research has been conducted in recent years related to developing and constructing different scales to measure different aspects of students’ attitudes, knowledge and competences related to sustainability and sustainability education. Some instruments, such as Tahmassebi and Najmi (2023), focussed on the assessment of educational practices regarding PRME in business schools. Another domain of research was the construction and validation of attitude scales to measure whether students have favorable attitudes regarding sustainability. Well-known sustainability attitude scales include the Environmental Paradigm (NEP) scale (Dunlap & Kent, 1978), the Environmental Attitudes of the University (EAU) scale (Fernández, Luis, & Jose, 2007), the Ecocentric and Anthropocentric Attitudes towards the Sustainable Development (EAATSD) (Kopnina & Cocis, 2017), the Children’s Environmental Attitude and Knowledge scale (CHEAKS) scale (Major et al., 2017), and the Attitude towards Sustainable Development scale (Biasutti & Frate, 2017). Other scales and instruments focused on determining students’ (interest in) sustainability knowledge and competences (see, for instance, Michalos et al. (2011), Muñoz, José, Fermín, Ángela, and Sara (2020), Manresa et al. (2021) and Hyttinen et al. (2023)). Research has also investigated sustainability attitudes specifically among business economics students and/or their attitudes towards corporate social responsibility (Doh and Tashman (2014), Kolodinsky et al. (2010) and Noguera-Méndez and Cifuentes-Faura (2022)). Within this domain, researchers investigated whether socio-demographic characteristics like gender, age and background influence students’ perceptions about whether they think businesses should pay attention to CSR/sustainability (Anand & Singh, 2021). What is less clear, however, is why a business should act sustainably among business economics students or what business economics students expect from a responsible businessperson. Students internalise a certain view on a business responsibilities to act responsibly during formal, non-formal and informal learning (Barth et al., 2007). The context, situation, and other factors influence their view of a responsible businessperson. Also, teachers internalise a certain view or perspective on how they see the responsibility of a businessperson. It is important to understand the influence of these views and to make them explicit because the pedagogical approaches used and the way teachers teach can depend on their own views and discourse (Kemper et al., 2019).

Against this background, the purpose of the present article was to develop, test and validate an instrument to detect the views that business economics students have internalised. The central research question in the paper is ‘How can students’ perspectives on the role of a responsible businessperson be assessed?’ To the best of the authors’ knowledge, no instruments previously existed to assess the perspectives of business economics students on the role of a business in relation to sustainability. This new instrument makes it possible to detect the starting point of both students and teachers related to their norms and values regarding sustainability in business and to investigate the extent to which those internalised perspectives might evolve as a consequence of interactions between students and teachers. In addition, changes in views regarding a responsible businessperson can indicate transformative learning. In contrast to previous work, which focused mainly on fixed characteristics like attitudes, knowledge and competences, the present research focuses on the views of students in regard to the role of a responsible businessperson. More specifically, the research instruments help to detect the route that students choose when they think about sustainability concerns and decisions in a business context and clarifies what expectations they respond to. Although the actions taken by a responsible businessperson can be the same, the motivation to act can differ. The instrument provides insights into the guiding principles to act for sustainability in business. These guiding principles are either external (that is, following external motivations) or internal (that is, creating new paths to sustainability without the guidance from predefined regulations). A transition by business economics students from external guiding principles to internal motivations to act for sustainability within business is an example of transformative learning in business economics education.

The remainder of this article is structured as follows. The next section will explain the development process of the instruments and the methods used to test the validity and reliability of the instrument. The instrument and the results of the validity and reliability test are presented in Section 3. Section 4 discusses the implications, limitations and recommendations for further research.

2. Methods

In this study, a mixed-method research design, as described by Zhou (2019), was used for scale development and validation. As illustrated in Fig. 1, the research consisted of three phases.

2.1. Construction of the pre-instrument

To define the construct and content domain of the instrument, literature on sustainability in business education and responsibilities within business was consulted. The goal was to achieve insight into what is already known about different perspectives and reasons why a businessperson should act sustainably or to define how a ‘responsible business’ is seen. What can be defined as sustainable business actions depends on how a business’ responsibilities in regard to sustainability are defined. We searched the literature to generate items that could be used in the instrument to measure the construct. Relevant literature was gathered from Web of Science and Google Scholar. To determine the search terms, the researchers discussed and selected the relevant search terms to gain knowledge about what is known in this field. The search terms ‘Responsible Business education’, ‘Sustainability AND (Business) Economics Education’, ‘Corporate Social Responsibility AND (Business) Economics Education’ were used. Next, snowballing methods were used to collect literature (Greenhalgh & Peacock, 2005). Subsequently, items were created based on a typology of responsibilities described by Pellizzoni (2004), the roles of a responsible businessperson as identified by Andersson (2016) and Raworth’s propositions on how to do business within the doughnut. The different items were discussed with the research team to ensure content validity. We had the opportunity to pre-test the questionnaire in collaboration with a local newspaper. However, given the tight deadlines imposed by the newspaper, the evaluation of the questions by the target population was impossible. To ensure face validity, colleagues of the

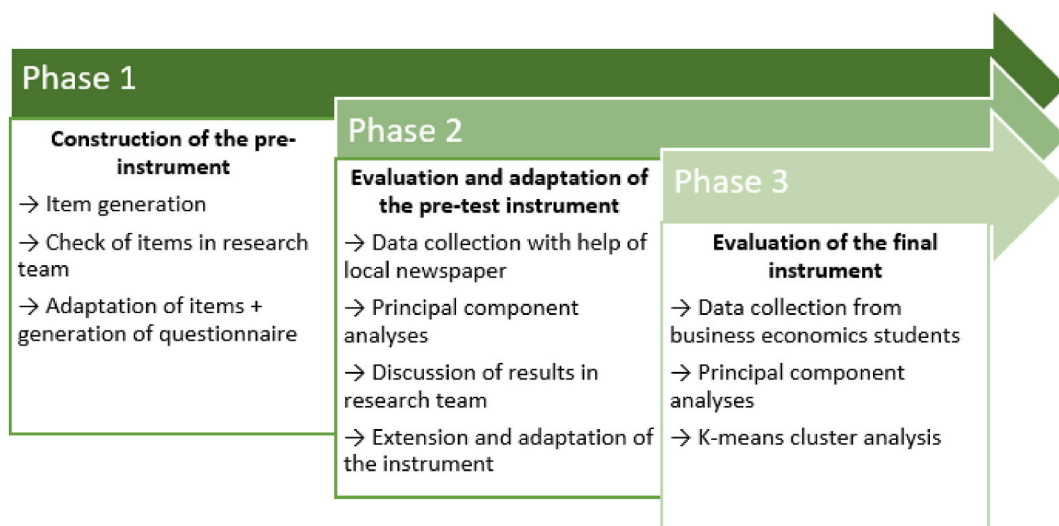


Fig. 1. Flow of the scale construction.

Table 1

Overview of courses that address sustainability per study program.

	Business administration	Applied Economics	Business engineer
1st bachelor	/	Sustainability-related topics are covered during Macroeconomics	Sustainability-related topics are covered during Macroeconomics
2nd bachelor	The topic of externalities is covered during Microeconomics	/	/
3rd bachelor	Corporate social responsibility	Sustainable innovation	Sustainability-related concepts are covered in the Technology Management course
Masters	Depending on major	Depending on major	Depending on major

researchers not involved in this study were asked to evaluate the different questions.

2.2. Evaluation and adaptation of the pre-test instrument

Next, pre-testing of the instrument was done. The literature on scale development recommended conducting different pilot test rounds (Zhou, 2019). The first test round was done with the help of a local newspaper because it provided the opportunity to collect a lot of data in a limited time. The first version of the instrument was spread in the province of Limburg in Belgium. The instrument was added in a questionnaire that investigated how important citizens find the topic of sustainability. The collaboration with the local newspaper represented an opportunity to collect a lot of data in a limited time frame, although this also implied a deficiency with regard to the sampling as the sample consisted of people with different backgrounds. In contrast, the sample should ideally consist of only business economics students. To control for careless responding, we followed Curran's (2016) suggestion to include a bogus check. Also, the discriminatory power of the instrument is checked by calculating the variance in answers for each respondent (Meade & Craig, 2012). After data cleaning, 4418 responses remained. Data were analysed using principal component analysis (PCA) and K-means cluster analysis. Principal component analysis was chosen because this method is most appropriate for identifying latent dimensions represented in the common variance of the variables (Hair, 2009). An exploratory approach was used to explore the underlying structure of the data because no scale or theoretical model exists and Hair (2009) advised an exploratory approach in early stages of instrument development. The theoretical model developed and tested was based on the work of Andersson (2016), Pellizzoni (2004) and Raworth (2017). To be able to conduct a PCA and K-means cluster analysis, we first checked some assumptions. Firstly, data have to be metric. Secondly, the sample needs to have at least five times as many observations as the number of items analysed with a minimum of 100 observations. Next, the Bartlett's test of sphericity, the Kaiser-Meyer-Olkin measure of sampling adequacy (KMO) and the measure of sampling adequacy (MSA), were computed to check whether the items have sufficient correlation. The threshold values of these different tests are a significant Bartlett's test of sphericity, a KMO larger than 0.6 and MSA values higher than 0.5. A PCA with Varimax rotation was computed to detect the structure of the items. Hereafter, the statements of the instrument were adapted because the results of the pre-test showed a low Cronbach's alpha. Furthermore, we added additional items to extend the instrument.

2.3. Evaluation of the final instrument

In the final phase, the new version of the instrument was tested again. This time, data were collected using voluntary response sampling. Students of the faculty of business economics of Hasselt University were invited by email to participate in the research. Students of all ages (that is, first-, second- and third-year bachelor and master students from the business administration, applied economics and business engineer study programmes were invited. The topic sustainability is limited covered in their courses (see Table 1 for an overview for each study programme of the courses where sustainability-related topics are covered).

The researchers reached out to students during their classes to increase response rates. In total, 1304 students were asked to participate in the research. Of these, 234 students filled in the questionnaire after having read the informed consent. After data cleaning, 35 answers were deleted because they were incomplete. The final sample consisted of 199 respondents. Next, an exploratory

Table 2
Theoretical model for the instrument.

	Pellizzoni's regimes	Andersson's roles	Raworth's approaches
Not included	Care •Fulfilling the needs of stakeholders (Business knows what is best) •Business as 'a good master'	<i>Could not be identified in the empirical studies.</i>	<i>Not described/mentioned</i>
Follow the law	Liability •Good governance •Responsibilities defined by law •Legal sanctioning •External moral driver	Adapting role •Follow the law •Maximise profit •No personal convictions	Do nothing •Follow the law •Maximise profit
Follow customers interest and make profit	Accountability •Governance •Voluntary actions •Responsibilities defined by customers •External moral driver	Adding role •Maximise profit •Respond to demands external to the business (customers)	Do what brings results •Maximise profit •Eco-efficient measures to lower costs •Eco-efficient measures to increase the popularity of the brand
Creating new paths to sustainability	Responsiveness •Sensitive to the needs of stakeholders •Internal moral driver	Creating role •Business to make change towards sustainability •Sensitive to the needs of all stakeholders	Deliver a fair share to sustainability and cause no harm •Do not cause harm •Try to have a positive impact

PCA and reliability tests were conducted to evaluate the instrument. IBM SPSS Statistics version 28 software was used to do the analysis. To be able to conduct a PCA and K-means cluster analysis, the Bartlett's test, KMO values and MSA values were checked before conducting the analyses. When doing the analysis, an orthogonal rotation method was chosen as the component correlation matrix showed the correlation between the items was too low for oblique rotation. Specifically, the varimax rotation was chosen and outcomes were sorted by size. Once the factor solution was final, Z-scores were calculated and a K-means cluster analysis was computed to search for the different 'roles of a responsible businessperson'.

3. Results

We first present the literature review results; that is, the theoretical model upon which the instrument was based. Next, the results regarding the pre-test instrument are presented; that is, its conception and the results obtained from including the instrument in the questionnaire of the local newspaper. Subsequently, we present how the instrument was improved based on those results. Next, the final instrument is proposed based on the results of the principal component analysis. Lastly, the final instrument is used to cluster business economics students in terms of their perspective of a businessperson in acting sustainably.

3.1. Theoretical basis

Literature on responsibilities and sustainability in business was analysed to generate items to construct the measurement instrument. Three studies found during the literature study were used as a theoretical basis upon which to construct the instrument. Table 2 provides an overview of the theoretical model generated based on the roles of a responsible businessperson from Andersson (2016), the approaches of Raworth (2017) on the integration of sustainability in business and the responsibility regimes of Pellizzoni (2004) (more details about the different models see introduction). The care regime of Pellizzoni (2004) was not included in the theoretical basis of this instrument because no comparison was found with Andersson (2016) and Raworth (2017).

The purpose of the measurement instrument is to detect students' perspectives on the role of a business in relation to sustainability. The guiding principle to integrate sustainability in business can differ between (1) external or amoral motivations by following the law or customers' interest in making a profit, or (2) internal or moral motivations by creating new paths to sustainability. In this way, the instrument contributes to making the assessment of transformative learning in business economics education possible. As defined in the introduction, transformative learning "(...) involves a deep structural shift in the basic premises of thought, feelings and actions. It is a shift of consciousness that dramatically and permanently alters our way of being in the world" (Morrell & O'Connor, 2002, p. 17). A shift from external or amoral guiding principles to internal and moral guiding principles to act for sustainability in business can be defined as 'transformative' because students change their worldview on why a business should act for sustainability; that is, a change in their perspective from 'following' to 'creating'.

3.2. Construction and evaluation of the pre-test instrument

The aim of this research was to develop an instrument that measures the construct of 'a student's view on the role of a responsible businessperson. The first phase of the research consisted of creating a topic list based on the model of Pellizzoni (2004), the roles of a responsible businessperson that Andersson (2016) identified empirically, and Raworth's (2017) approaches to doing business sustainably. The topic list was the input for formulating statements with which respondents could agree or disagree. The statements were created and discussed together with the research team. As the instrument was part of a bigger questionnaire from a local newspaper, the number of statements the researchers could include was limited to eight. The respondents were asked whether or not they agreed with the statements, using a five-point Likert scale with one representing fully disagree and five representing fully agree. The

Table 3

First version of the instrument translated from Dutch to English to detect a person's view on the roles of a responsible businessperson.

Statements	Follow customers interest and make profit	Follow the law	Creating new paths to sustainability
1. A businessperson should focus on making profits even if this makes the business unsustainable.	x		
2. Laws and regulations should ensure businesses work in a sustainable way, business should not do this on their own accord.		x	
3. A businessperson may take into account personal convictions when making a business decision.			x
4. A businessperson has to produce what consumers want, even when consumers ask for unsustainable products.	x		
5. Businesses should lead by example and help the rest of society to be more sustainable.			x
6. Not making profit but making society more sustainable should be a business's main goal.			x
7. A business should only take responsibility for sustainability when required by laws and regulations.		x	
8. A businessperson needs to be sensitive to the needs of all stakeholders.			x

statements of this instrument are presented in Table 3.

In Phase 2, the instrument presented in Table 3 was tested. Thanks to the collaboration with the newspaper, the researchers collected 4418 responses. First, the assumptions as described in the method section were checked. The data of the sample are metric and the sample size is sufficient to conduct a PCA. The Bartlett's test of sphericity, the KMO and the MSA meet the threshold values. The significant Bartlett's test of sphericity indicates that there is significant correlation for at least some variables. The KMO test shows that the dataset is suitable for PCA, as there is common variance among the variables. The MSA values are an additional check to show the degree of intercorrelations among the variables (see Table 4).

Next, a PCA analysis with varimax rotation was conducted (see the results in Annex A). This resulted in a three-factor solution that explained 57.79% of the variance. It was not possible to calculate a Cronbach's alpha for each factor because of the low number of items in each factor. Therefore, we calculated the Cronbach's alpha of the entire model. The value of 0.361 was too low, according to Hair (2009). The pre-test instrument was insufficient to measure the defined construct, so it made no sense to perform subsequent PCA analysis and K-means cluster analysis.

3.3. Adaptation and evaluation of the instrument

The limited number of statements probably explains why the statements in the pre-test were incapable of measuring the predefined construct. Therefore, the scale was reconstructed and expanded with more items. See the new version in Table 5.

3.4. Evaluation of the final instrument

In Phase 3, the new version of the instrument (see Table 5) was tested. Data were collected from students of the faculty of business economics at Hasselt University. The researchers chose to work with the target population of the instrument in the sample for this second round. Although the collaboration with the newspaper was an opportunity to generate a bigger sample, it did not consist of only participants with a business economics background. Also, a collaboration with the newspaper in this second round was not possible. To check the discriminatory power of the instrument, the variance in answers for each respondent was checked (Meade & Craig, 2012). No observations were found with a variance close to or equaling 0, indicating no careless responding. The same assumptions regarding data suitability as described in the method section were checked before the start of the analyses. The data are metric and the sample size of 199 complete observations exceeds the minimum requirement of five times the number of items analysed. The MSA values, the Bartlett's test of sphericity and the KMO value meet the threshold value (see Table 6). The Bartlett's test of sphericity reveals a notable correlation among certain variables. The KMO test affirms the dataset's suitability for PCA, signifying shared variance among the variables. Additionally, MSA values serve as an extra measure, indicating the extent of intercorrelations among the variables.

An exploratory PCA with Varimax rotation was performed. The varimax rotation was chosen because none of the correlations of the component correlation matrix exceeded the value of 0.32, which indicates that no too high correlation exists between the components (Tabachnick, Fidell, and Ullman 2007). The items are sorted by size and only items with a communality exceeding 0.40 were kept in the PCA. In a sample of 199 respondents, factor loadings higher than or equal to 0.40 are considered to identify significant factor loadings (Hair, 2009). This means that items 2, 4, 6, 8, 18, 19 and 21 from Table 3 were deleted, because the communality of those variables was too low. Moreover, Item 10 was deleted based on the results of the reliability tests because the 'Cronbach's alpha if item deleted' was extremely high for Item 10. The final factor solution consists of 13 items across four factors, explaining 60.39% of the total variance. However, in this factor solution, Item 3 – 'It's up to government to demand more sustainability by regulation, not to a businessperson' – is a factor on its own with cross-loadings to the other factors. We deleted Item 3 because interpretation compared to the other factors is not possible and then conducted the PCA analysis and reliability tests again. The final factor solution (see Table 7) consists of 12 items divided among three factors, which explains 55.05% of the total variance. The factor solution presented in Table 7 is theoretically explicable and has a Cronbach's alpha of 0.699. To control the outcome of the final factor solution, the dataset was split randomly into two equal samples. For each of these samples, a confirmatory principal component analysis was runned to check whether the outcome was consistent. The same factor solution was found for both samples as in the initial solution.

Items 5, 7, 9, 12, 13, 14, 17 and 20 are grouped together as one factor. This factor is defined as 'following an external driver' because all the statements indicate that 'acting responsibly as a business' is only necessary if somebody, something or some institution external to the business asks for it. Compared to the conception of the scale in Table 2, the statements that group as the factor 'following an external driver' are the ones classified as 'follow law' and 'follow customers' interest and making profit. Items 1 and 11 were grouped together as one factor, which the researchers defined as 'profit'. This name is chosen because both statements in this factor state that the main responsibility of a business is to generate profit. Items 15 and 16 can be grouped as a factor defined as 'internally driven.' Both statements reflect the concept in the category 'creating new paths to sustainability of the theoretical model in Table 2.'

Table 4
Results of the KMO and Bartlett's test – pre-test instrument.

KMO and Bartlett's test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy		0.699
	Approx. Chi-Square	3815.176
Bartlett's Test of Sphericity	df	28
	Sig.	<0.000

Table 5
New version of the instrument.

Statements	Follow customers interest and make profit	Follow the law	Creating new paths to sustainability
1. A businessperson should focus on making profits.	x		
2. A businessperson should feel responsible for improving society instead of making profits.			x
3. It's up to government to demand more sustainability by regulation, not to a business person.		x	
4. A businessperson has the most influence on sustainability and they, more than others, should lead the way to a more sustainable future.			x
5. A businessperson should only take responsibility for the environment if there are laws protecting the environment.		x	
6. A businessperson should never rely on personal convictions when making a business decision.		x	
7. A businessperson has to produce what consumers want, even when customers ask for unsustainable products.	x		
8. A businessperson needs to be sensitive to the needs of all stakeholders.			x
9. A businessperson should only take responsibility for sustainability if consumers demand sustainable products.	x		
10. A businessperson should proactively try to improve society and not wait for consumers to demand it.			x
11. A businessperson should mainly think about making profit but try to be sustainable while doing so.	x		
12. A businessperson should only take responsibility for sustainability if they could get eco-certification for this.	x		
13. A businessperson can pay more attention to sustainability, only if laws and regulations from the government request more sustainable actions from businesses.		x	
14. If a businessperson wants to take responsibility for sustainability, they should only do what is requested to get a green label.	x		
15. When making a business decision, a businessperson needs to consider the social and environmental impact first.			x
16. A businessperson needs to be sensitive to the needs of nature, including animals.			x
17. A businessperson should choose the cheapest option to produce goods and services, even if this means that production should take place in regions where working conditions are worse than our standards.		x	
18. A businessperson should consider shareholder value as the result of their work, not their goal.			x
19. A businessperson should feel responsible for all consequences, whether intended or not.			x
20. A businessperson should only act for more sustainability when the government imposes laws and regulation to do so.		x	
21. A businessperson's job is not only to make a business do well, but also to make it do good.			x

Table 6
Results of the KMO and Bartlett's test – pre-test instrument.

KMO and Bartlett's test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy		0.781
Approx. Chi-Square		571.74
Bartlett's Test of df		66
Sphericity Sig.		<0.001

After the PCA, we conducted a K-means cluster analysis to identify whether the business economics students could be clustered in specific groups. First, the number of clusters was fixed at 3 as the theoretical model predicts three different roles: follow the law, follow customers and make profit and creating new paths to sustainability. However, a cluster solution of two clusters is also possible as the most important distinction between the different roles is the moral driver, which is external when following the law or following customers and make profit and internal when creating new paths to sustainability. Therefore, a K-means cluster analysis was also conducted with the number of clusters fixed at two. When comparing the three-cluster solution with the two-cluster solution, only the two-cluster solution was theoretically explicable. The two-cluster solution is presented in Fig. 2. Cluster 1 contains 123 respondents. The respondents of Cluster 1 seem to perceive the role of a businessperson as corresponding to creating new paths of sustainability. This is illustrated by a positive loading on Items 15 and 16. All other statements are negatively rated by the respondents in this cluster. This shows that the respondents of Cluster 1 agree that a businessperson needs to be driven by an internal moral driver to act responsibly towards all stakeholders. Therefore, Cluster 1 is named 'Creator'. This can be linked to the creating role of Andersson (2016) and deliver a fair share to sustainability and cause no harm of Raworth (2017). Cluster 2 contains 76 respondents and perceives the role of a businessperson as doing what is required by laws and regulations. This is illustrated by a positive loading on Items 5, 13

Table 7
Rotated component matrix – PCA varimax with Kaiser normalization.

Rotated Component Matrix	Following an external driver	Profit	Internally driven
A businessperson should only take responsibility if they could get eco-certification for this. (12)	0.748		
If a businessperson wants to take responsibility, they should only do what is requested to get a green label. (14)	0.722		
A businessperson can pay more attention to sustainability, only if laws and regulations from the government request more sustainable actions from businesses. (13)	0.697		
A businessperson should only take responsibility for the environment if there are laws protecting the environment. (5)	0.683		
A businessperson should choose the cheapest option to produce goods and services, even if this means that production should take place in regions where working conditions are worse compared to our standards. (17)	0.679		
A businessperson should only act for more sustainability when the government imposes laws and regulation to do so. (20)	0.647		
A businessperson should only take responsibility for sustainability if consumers demand sustainable products. (9)	0.603		
A businessperson has to produce what consumers want, even when customers ask for unsustainable products. (7)	0.508	0.41	
A businessperson should mainly think about making profit but while doing so, try to be sustainable. (11)		0.823	
A businessperson should focus on making profits. (1)		0.788	
A businessperson needs to be sensitive to the needs of nature including animals. (16)			0.812
When taking a business decision, a businessperson needs to think first about the social and environmental impact that the decision creates. (15)			0.797

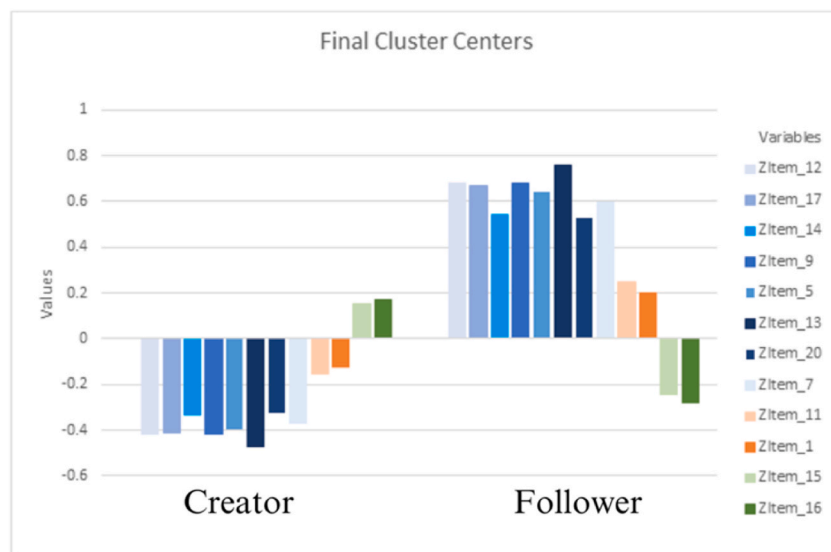


Fig. 2. Results K-means cluster analysis.

and 20. Furthermore, the respondents in this cluster also see the responsibility of a businessperson as following what consumers want and to make profit as illustrated by a positive loading on Items 7, 9 12 and 14. Respondents in this category do not agree with Items 15 and 16, while they agree with all other statements. The respondents in Cluster 2 are the opposite of those in Cluster 1. They think a businessperson should only act like a 'responsible' businessperson if they are externally motivated to do so. Therefore, cluster 2 is named 'Follower'. The respondents' view in Cluster 2 corresponds to the adapting role and adding role of [Andersson \(2016\)](#) and 'do nothing' or 'do what brings results' from [Raworth \(2017\)](#). The results of the ANOVA table (see Annex B) prove that the two clusters are significantly different on a 95 per cent confidence level.

4. Discussion and conclusion

4.1. Findings

The results of the study present an instrument that assessed students' perspectives on the role of business with regard to

sustainability, and divided the respondents into two different clusters. Within the first cluster, named *creator*, respondents perceive the responsibility of a businessperson as being sensitive to the needs of all stakeholders and the broader society. The guiding principle to act for sustainability is internal. Business is seen as way of creating new paths to sustainability. Respondents in this group believe that a businessperson should proactively strive for social and environmental sustainability. The perspective of the respondents in Group 1 corresponds to [Andersson's \(2016\)](#) creating role and [Raworth's \(2017\)](#) approach of delivering a fair share to sustainability and causing no harm. Respondents in Cluster 2 perceive the responsibility of a responsible businessperson as following the rules and laws of the government and meeting the requirements of customers while also making profit. There is no need to act towards sustainability proactively. The motivation to act responsibly as a businessperson is driven externally by customers, profit and/or the law. Therefore, Cluster 2 is named '*follower*'. The perspective of the respondents in Cluster 2 corresponds to a combination of the adapting and adding roles, as defined by [Andersson \(2016\)](#), and the approaches of 'do nothing' and 'do what brings results' of [Raworth \(2017\)](#). Specifically, at Hasselt University, 62% of the business economics students who participated in the research are classified in the *creator* cluster, whereas 38% are classified in the *follower* cluster. This means that 62% of the students felt that businesses should do more regarding sustainability than what is requested by law, legislation or customers.

4.2. Theoretical and practical contributions

There is growing interest in integrating sustainability, ethics and responsibilities in business economics education ([Lozano et al., 2013](#); [Rusinko & Sama, 2009](#)). Business economics education as a discipline has a remarkable influence in achieving sustainable development ([Ceulemans et al., 2011](#); [Vargas-Merino et al., 2024](#)). Business schools worldwide have embraced the Principles for Responsible Management Education (PRME) ([Millar & Price, 2018](#)). The PRME, an initiative of the United Nations, aims to stimulate a fundamental change in business economics education to achieve the Agenda 2030 and reach the sustainable development goals (SDGs) ([Storey et al., 2017](#)). Despite increasing attention, a fundamental change in business economics education remains lacking ([Russo et al., 2023](#)). The implementation of sustainability and PRME is complex and the topic should be addressed holistically.

New teaching approaches based on active learning principles like transformative learning are encouraged because they can enable fundamental changes ([Vargas-Merino et al., 2024](#)). However, little is known about how transformative learning for sustainable development is addressed in business economics education. Only a few case studies exist; for example, [Brunstein, Walvoord, and Cunliff \(2021\)](#) and [Corazza et al. \(2022\)](#). The development of instruments to assess transformative learning makes the recognition of best practices and good examples easier. However, the assessment of transformative learning is challenging as the change process is internal and a transformation requires a longitudinal approach ([King, 2021](#)). A change from one view of the role of a responsible businessperson to another might indicate a transformative learning experience in the field of business economics education. As deduced from the definition provided in the introduction, the transformative learning process 'involves a deep structural shift in thoughts feelings and actions' ([Morrell & O'Connor, 2002](#), p.17). Translated to business economics education, an example of such a change of perspective can be from 'following' as a guiding principle to act for sustainability in business to 'creating'. Although actions from businesspeople with different perspectives might be the same, the internal moral driver or guiding principle to act is different. This difference in guiding principle is important and can create fundamental implications. When the guiding principle to act for sustainability is 'following', the dependence on 'good' laws and 'interested' consumers is critical. Indeed, if laws and customers do not call for sustainability, nothing will happen. In addition, 'following' implies that laws and consumers know how to address the complex sustainable issues, which is often not the case. Therefore, to reach sustainability and to address the complexity of sustainability issues, the authors agree with [Andersson \(2016\)](#), [Pellizzoni \(2004\)](#) and [Raworth \(2017\)](#) that a 'creator' perspective is preferred. The advantage of the developed instrument is that it is an easy and efficient way to achieve insights into the perspectives regarding the responsibility of a businessperson towards sustainability. This can help teachers and researchers discover the starting point of students related to the norms and values they have regarding business responsibilities. It allows teachers to differentiate in their teaching approach and select the appropriate learning content and strategies. Next, the instrument can also be useful for teachers and researchers to investigate whether a transition happened in the values, norms and views of students. In follow-up research, this instrument can be used to investigate which didactical formats trigger the transformative learning process, as this is an easy and efficient way to have an initial indication of whether transformation is happening. To gain additional insight in the process of transformative learning itself, researchers can combine the use of this instrument with additional qualitative research to investigate the change in perspectives. In this way, formative and summative assessment are connected and could work in harmony to improve our understanding of the transformative learning process ([Lau, 2016](#)).

4.3. Limitations and perspectives for future research

Although considerable attention has been paid to the development of the instrument, certain limitations need to be considered. Firstly, our aim was to develop an instrument that would be used to detect the role regarding a responsible businessperson as perceived by business economics students. However, the data of the pre-test are not limited to business economics students only. For the first test round, data were gathered with the help of a local newspaper because this provided the opportunity to collect a lot of data in a limited time frame. However, the data collection during the second test round was done by business economics students. Moreover, the final test sample only involved business economics students from one university. Therefore, testing the instrument in other regions is recommended in order to check for possible regional effects. Respondents were invited by email and personal appeal during classes to participate in a survey about sustainability in business. However, they were free to choose whether they wanted to participate, which could create some self-selection bias. Ideally, a random selection of participants will be made, although this is not always possible in

educational research for practical and ethical reasons (Pring, 2010). Next, data are collected by self-reporting questionnaires, leaving a possibility of socially desirable answers. However, the researchers tried to limit this bias by being ‘vague’ about the research goal in the invitation. Future research can focus on further development of instruments to assess transformative learning for sustainable development in business economics education. The instrument presented is to the best of the authors’ knowledge the first to make the assessment of transformative learning for sustainable development possible specifically in the field of business economics education. As different types of transformation can occur in the business economics class, other tools to detect ‘other types’ of transformation are necessary. The development of the same type of instruments in other disciplines could also be valuable because to transform to a sustainable society, sustainability and responsibility should be addressed in every discipline. Longitudinal studies are required to investigate of transformative learning creates (behavioural) change in practice.

Ethical approval

The researchers received approval from SMEC UHasselt (Reference REC/SMEC/2021-22/21).

CRediT authorship contribution statement

Lise Janssens: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft. **Tom Kuppens:** Conceptualization, Supervision, Writing – original draft, Writing – review & editing. **Pernilla Andersson:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Katrien Struyven:** Conceptualization, Writing – original draft, Writing – review & editing.

Declaration of interest

None.

Data availability

The data that has been used is confidential.

Annex A – Results factor analysis and cluster analysis pre-test

Rotated Component Matrix			
	Factor 1	Factor 2	Factor 3
A business person should focus on making profits even if this makes the business unsustainable. (Q30_1)	-0.304	0.470	0.456
Laws and regulation should ensure business work in a sustainable way, business should not do this on their own accord. (Q30_2)		0.706	
A business person may take into account personal convictions when making a business decision. (Q30_3)			0.86
A business person has to produce what consumers want, even when consumers ask for unsustainable products. (Q30_4)		0.503	0.515
Businesses should lead by example and help the rest of society to be more sustainable. (Q30_5)	0.758		
Not making profit but making society more sustainable should be a business's main goal. (Q30_6)	0.685		
A business should only take responsibility for sustainability when required by laws and regulation. (Q30_7)		0.747	
A business person needs to be sensitive to the needs of all stakeholders. (Q30_8)	0.671		

Annex B – ANOVA table cluster solution

ANOVA						
	Cluster		Error		F	Sig.
	Mean Square	df	Mean Square	df		
ZItem_12	56.801	1	0.717	197	79.249	0.000
ZItem_17	54.444	1	0.729	197	74.713	0.000
ZItem_14	36.402	1	0.820	197	44.377	0.000
ZItem_9	56.458	1	0.718	197	78.579	0.000
ZItem_5	50.969	1	0.746	197	68.291	0.000
ZItem_13	71.213	1	0.644	197	110.650	0.000
ZItem_20	33.984	1	0.833	197	40.819	0.000
ZItem_7	43.479	1	0.784	197	55.432	0.000
ZItem_11	7.474	1	0.967	197	7.728	0.006
ZItem_1	4.965	1	0.980	197	5.066	0.025
ZItem_15	7.426	1	0.967	197	7.676	0.006
ZItem_16	9.433	1	0.957	197	9.855	0.002

References

- Anand, M., & Singh, J. (2021). Business students' perception of corporate social responsibility: An exploratory study. *Decision*, 48, 261–284.
- Andersson, P. (2016). The responsible business person: Studies of business education for sustainability. *Södertörns högskola*.
- Andersson, P. (2018). Talking about sustainability issues when teaching business economics—The 'Positioning' of a responsible business person in classroom practice. *Journal of Social Science Education*, 17(3), 46–62.
- Annan-Diab, F., & Carolina, M. (2017). Interdisciplinarity: Practical approach to advancing education for sustainability and for the sustainable development goals. *International Journal of Management in Education*, 15(2, Part B), 73–83.
- Backhouse, R. E., & Medema, S. G. (2009). Retrospectives: On the definition of economics. *The Journal of Economic Perspectives*, 23(1), 221–233.
- Barth, M., Godemann, J., Rieckmann, M., & Stoltenberg, U. (2007). Developing key competencies for sustainable development in higher education. *International Journal of Sustainability in Higher Education*.
- Baumgartner, R. J. (2014). Managing corporate sustainability and CSR: A conceptual framework combining values, strategies and instruments contributing to sustainable development. *Corporate Social Responsibility and Environmental Management*, 21(5), 258–271.
- Beddewela, E., Warin, C., Hesselden, F., & Coslet, A. (2017). Embedding responsible management education – staff, student and institutional perspectives. *International Journal of Management in Education*, 15(2, Part B), 263–279.
- Bianchi, G., Pisiotis, U., & Cabrera Giraldez, M. (2022). GreenComp the European sustainability competence framework. *Joint Research Centre (Seville site)*.
- Biasutti, M., & Frate, S. (2017). A validity and reliability study of the attitudes toward sustainable development scale. *Environmental Education Research*, 23(2), 214–230.
- Birks, S. (2012). Rethinking economics: Theory as rhetoric. *Bristol Business School Economics Papers Series*, 12, 13.
- Brunstein, J., Mark Edward, W., & Ed, C. (2021). Problem-posing in management classrooms for collective sustainability transformation. *International Journal of Sustainability in Higher Education*, 22(3), 477–496.
- Ceulemans, K., De Prins, M., Cappuyns, V., & De Coninck, W. (2011). Integration of sustainable development in higher education's curricula of applied economics: Large-scale assessments, integration strategies and barriers. *Journal of Management and Organization*, 17(5), 621–640.
- Colander, D. (2017). Integrating normative issues in the principles of economics texts: Introduction to a symposium. *The Journal of Economic Education*, 48(1), 1–2.
- Colander, D., Goldberg, M., Haas, A., Juselius, K., Kirman, A., Lux, T., & Sloth, B. (2009). The financial crisis and the systemic failure of the economics profession. *Critical Review*, 21(2–3), 249–267.
- Corazza, L., Cottafava, D., & Torchia, D. (2022). Education for sustainable development: A critical reflexive discourse on a transformative learning activity for business students. *Environment, Development and Sustainability*.
- Curran, P. G. (2016). Methods for the detection of carelessly invalid responses in survey data. *Journal of Experimental Social Psychology*, 66, 4–19.
- Darity, W. (2022). Alternatives to the scarcity principle. *The Journal of Economic Education*, 53(4), 340–347.
- de Muijnck, S., & Tieleman, J. (2022). *Economy studies: A guide to rethinking economics education*. Amsterdam University Press.
- Doh, J. P., & Tashman, P. (2014). Half a world away: The integration and assimilation of corporate social responsibility, sustainability, and sustainable development in business school curricula. *Corporate Social Responsibility and Environmental Management*, 21(3), 131–142.
- Dunlap, R. E., & Kent, D. Van L. (1978). The "new environmental Paradigm". *The Journal of Environmental Education*, 9(4), 10–19.
- Faure, M. G. (2009). Environmental liability. *Tort law and economics* (286). Edward Elgar.
- Fernández, M. R., Luis, R.-B., & Jose, C. (2007). Evaluation of environmental attitudes: Analysis and results of a scale applied to university students. *Science Education*, 91(6), 988–1009.
- García-Feijoo, M., Eizaguirre, A., & Rica-Aspiunza, A. (2020). Systematic review of sustainable-development-goal deployment in business schools. *Sustainability*, 12(1), 440.
- García-Rosell, J.-C. (2019). A discursive perspective on corporate social responsibility education: A story Co-creation exercise. *Journal of Business Ethics*, 154(4), 1019–1032.
- Gómez, F. U., Sáez-Navarrete, C., Lioi, S. R., & Marzuca, V. I. (2015). Adaptable model for assessing sustainability in higher education. *Journal of Cleaner Production*, 107, 475–485.
- Green, T. L. (2015). Lecturers' perspectives on how introductory economic courses address sustainability. *International Journal of Sustainability in Higher Education*.
- Greenberg, D. N., Deets, S., Erzurumlu, S., Hunt, J., Manwaring, M., Rodgers, V., & Swanson, E. (2017). Signing to living PRME: Learning from a journey towards responsible management education. *International Journal of Management in Education*, 15(2), 205–218. Part B.

- Greenhalgh, T., & Peacock, R. (2005). Effectiveness and efficiency of search methods in systematic reviews of complex evidence: Audit of primary sources. *BMJ*, 331 (7524), 1064–1065.
- Hair, J. F. (2009). *Multivariate data analysis*.
- Harris, S. M. (2007). Does sustainability sell? Market responses to sustainability certification. *Management of Environmental Quality: An International Journal*, 18(1), 50–60.
- Horton, R. V. (1972). Values and the economics principles courses: Values and knowledge. *The Journal of Economic Education*, 3(2), 118.
- Howlett, C., Ferreira, J.-A., & Blomfield, J. (2016). Teaching sustainable development in higher education: Building critical, reflective thinkers through an interdisciplinary approach. *International Journal of Sustainability in Higher Education*.
- Hyytiäinen, H., Laakso, S., Pietiäinen, J., Ratvio, R., Ruippo, L., Tuononen, T., & Vainio, A. (2023). Perceived interest in learning sustainability competencies among higher education students. *International Journal of Sustainability in Higher Education*, 24(9), 118–137.
- Kane, T. E. (2011). *Transforming the soul of education: Sustainability at the center of teaching and learning in secondary schools*. Portland State University.
- Kemper, J. A., Ballantine, P. W., & Hall, C. M. (2019). Combining the 'why' and 'how' of teaching sustainability: The case of the business school academics. *Environmental Education Research*, 25(12), 1751–1774.
- King, J. (2021). Assessing students in a transformative learning program. *Transformative Language Learning and Teaching*, 238–249.
- Kolb, M., Fröhlich, L., & Schmidpeter, R. (2017). Implementing sustainability as the new normal: Responsible management education – from a private business school's perspective. *International Journal of Management in Education*, 15(2, Part B), 280–292.
- Kolodinsky, R. W., Madden, T. M., Zisk, D. S., & Henkel, E. T. (2010). Attitudes about corporate social responsibility: Business student predictors. *Journal of Business Ethics*, 91, 167–181.
- Kopnina, H., & Cocis, A. (2017). Environmental education: Reflecting on application of environmental attitudes measuring scale in higher education students. *Education Sciences*, 7(3), 69.
- Lambrechts, W., Van Liedekerke, L., & Petegem, P. Van (2018). Higher education for sustainable development in Flanders: Balancing between normative and transformative approaches. *Environmental Education Research*, 24(9), 1284–1300.
- Lau, A. M. S. (2016). 'Formative good, summative bad?'—A review of the dichotomy in assessment literature. *Journal of Further and Higher Education*, 40(4), 509–525.
- Lozano, R., Lozano, F. J., Mulder, K., Huisingh, D., & Waas, T. (2013). *Advancing higher education for sustainable development: International insights and critical reflections*. Elsevier.
- Major, L., Namestovski, Ž., Horák, R., Bagány, Á., & Krekić, V. P. (2017). Teach it to sustain it! Environmental attitudes of Hungarian teacher training students in Serbia. *Journal of Cleaner Production*, 154, 255–268.
- Manresa, A., Berbegal-Mirabent, J., Faura-Martínez, Ú., & Llinares-Ciscar, J.-V. (2021). What do freshmen know about sustainability? Analysing the skill gap among university business administration students. *Sustainability*, 13(16), 8813.
- Maxfield, S. (2011). Teaching economics to business students through the lens of corporate social responsibility and sustainability. *The Journal of Economic Education*, 42(1), 60–69.
- McFarlane, D. A., & Ogazon, A. G. (2011). The challenges of sustainability education. *Journal of Multidisciplinary Research*, 3(3).
- Meade, A. W., & Craig, S. B. (2012). Identifying careless responses in survey data. *Psychological Methods*, 17(3), 437.
- Michalos, A. C., Creech, H., McDonald, C., & Kahlke, P. (2011). Knowledge, attitudes and behaviours. Concerning education for sustainable development: Two exploratory studies. *Social Indicators Research*, 100(3), 391–413.
- Millar, J., & Price, M. (2018). Imagining management education: A critique of the contribution of the United Nations PRME to critical reflexivity and rethinking management education. *Management Learning*, 49(3), 346–362.
- Morrell, A., & O'Connor, M. A. (2002). Introduction. In E. O'Sullivan, A. Morrell, & M. A. O'Connor (Eds.), *Expanding the boundaries of transformative learning: Essays on theory and praxis* (pp. 15–20). New York: Palgrave.
- Muñoz, R., José, M., Fermín, S.-C., Ángela, B.-R., & Sara, S.-G. (2020). Are we training in sustainability in higher education? Case study: Education degrees at the university of Salamanca. *Sustainability*, 12(11), 4421.
- Noguera-Méndez, P., & Cifuentes-Faura, J. (2022). Environmental sustainability in economics teaching: Analysing Spanish upper secondary economics textbooks. *Environmental Education Research*, 1–18.
- Pellizzoni, L. (2004). Responsibility and environmental governance. *Environmental Politics*, 13(3), 541–565.
- Pring, R. (2010). In Third (Ed.). *Philosophy of educational research* (ed. Vol. Book, Whole, Vol. 2, p. 2nd). United Kingdom: Bloomsbury Publishing.
- Raworth, K. (2017). *Doughnut economics: Seven ways to think like a 21st-century economist*. Chelsea Green Publishing.
- Romano, A. (2018). Transformative learning: A review of the assessment tools. *Journal of Transformative Learning*, 5(1).
- Rusinko, C. A., & Sama, L. M. (2009). Greening and sustainability across the management curriculum: An extended journey. *Journal of Management Education*, 33(3), 271–275.
- Russo, F., Anita, L. W., Shrestha, A., & Saratchandra, M. (2023). Responsible management education in business schools—high on principles but low on action: A systematic literature review. *International Journal of Management in Education*, 21(3), Article 100843.
- Saha, A. (2014). Corporate social responsibility and managerial competencies: A dynamic requirement. *10th European conference on management leadership and governance (ECMLG)*, VERN univ appl sci, Zagreb, Croatia, Nov 13–14.
- Seatter, C. S., & Kim, C. (2017). Teaching sustainability in higher education: Pedagogical styles that make a difference. *Canadian Journal of Higher Education*, 47(2), 47–70.
- Sidiropoulos, E. (2014). Education for sustainability in business education programs: A question of value. *Journal of Cleaner Production*, 85, 472–487.
- Sterling, S. (2011). Transformative learning and sustainability: Sketching the conceptual ground. *Learning and Teaching in Higher Education*, 5(11), 17–33.
- Storey, M., Killian, S., & O'Regan, P. (2017). Responsible management education: Mapping the field in the context of the SDGs. *International Journal of Management in Education*, 15(2), 93–103.
- Stough, T., Kim, C., & Cappuyns, V. (2021). Unlocking the potential of broad, horizontal curricular assessments for ethics, responsibility and sustainability in business and economics higher education. *Assessment & Evaluation in Higher Education*, 46(2), 297–311.
- Tabachnick, B. G., Linda, S. F., & Ullman, J. B. (2007). *Using multivariate statistics* (Vol. 5). MA: Pearson Boston.
- Tahmassebi, H., & Najmi, M. (2023). Developing a comprehensive assessment tool for responsible management education in business schools. *International Journal of Management in Education*, 21(3), Article 100874.
- UNESCO. (2021). Berlin declaration on education for sustainable development. *UNESCO world conference on education for sustainable development*. Berlin: UNESCO.
- Vargas-Merino, J. A., Rios-Lama, C. A., & Panez-Bendezú, M. H. (2024). Critical implications of education for sustainable development in HEIs—A systematic review through the lens of the business science literature. *International Journal of Management in Education*, 22(1), Article 100904.
- Wang, L., Malhotra, D., & Murnighan, J. K. (2011). Economics education and greed. *The Academy of Management Learning and Education*, 10(4), 643–660.
- Wiek, A., Bernstein, M. J., Laubichler, M., Caniglia, G., Minter, B., & Lang, D. J. (2013). A global classroom for international sustainability education. *Creative Education*, 4(4), 19.
- Zhou, Y. (2019). A mixed methods model of scale development and validation analysis. *Measurement: Interdisciplinary Research and Perspectives*, 17(1), 38–47.